

## MILINGIMBI, Australia

WMO: 941400

Lat: 12.0931S

Lon: 134.8919E

Elev: 17

StdP: 101.13

Time Zone: 9.50 (AUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 15.4       | 17.2 | 9.5                           | 7.4 | 21.6 | 11.3 | 8.4 | 22.5 | 8.2                   | 27.5 | 7.5 | 27.3 | 1.7                   | 160  | 0.370 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 12            | 6.7                    | 34.2            | 25.9 | 33.5 | 25.8 | 32.9 | 25.8 | 28.3                | 31.3 | 27.8 | 30.8 | 27.3 | 30.6 | 4.2                  | 350  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.6                            | 23.6 | 30.2 | 27.1 | 22.8 | 29.7 | 26.5 | 22.1 | 29.2 | 91.6          | 31.3 | 89.2 | 31.0 | 86.9 | 30.7 | 30.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 7.3           | 6.5  | 5.9 | 14.1                       | 36.0 | 1.7                | 1.2 | 12.9                                               | 36.9 | 11.9       | 37.6 | 11.0       | 38.3 | 9.8        | 39.2 |    |    |
| (5)               |      |     | 12.4                       | 28.2 | 1.5                | 0.6 | 11.3                                               | 28.6 | 10.5       | 28.9 | 9.7        | 29.2 | 8.6        | 29.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 27.0   | 28.6  | 28.3  | 28.3  | 27.8  | 26.7  | 24.9  | 24.2  | 24.2  | 25.8  | 27.3  | 28.7  | 29.1 |
|                                                                                  | DBStd       | 2.43   | 1.37  | 1.21  | 1.11  | 0.98  | 1.34  | 1.82  | 1.56  | 2.05  | 1.42  | 2.14  | 1.73  | 2.84 |
|                                                                                  | HDD10.0     | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1    |
|                                                                                  | HDD18.3     | 6      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 0     | 0     | 0     | 4    |
|                                                                                  | CDD10.0     | 6197   | 576   | 513   | 567   | 533   | 517   | 446   | 440   | 439   | 475   | 536   | 560   | 593  |
|                                                                                  | CDD18.3     | 3161   | 318   | 280   | 309   | 283   | 259   | 196   | 182   | 183   | 225   | 278   | 310   | 338  |
|                                                                                  | CDH23.3     | 34003  | 3699  | 3140  | 3438  | 3094  | 2531  | 1579  | 1379  | 1511  | 2173  | 3203  | 3895  | 4362 |
| CDH26.7                                                                          | 12831       | 1486   | 1166  | 1283  | 1157  | 866   | 439   | 339   | 401   | 668   | 1228  | 1742  | 2056  |      |
| Wind                                                                             | WSAvg       | 3.0    | 2.9   | 2.9   | 2.5   | 2.8   | 3.4   | 3.6   | 3.5   | 3.3   | 3.0   | 2.9   | 2.6   | 2.7  |
| Precipitation                                                                    | PrecAvg     | 1199   | 283   | 259   | 288   | 88    | 29    | 8     | 2     | 1     | 1     | 11    | 58    | 176  |
|                                                                                  | PrecMax     | 1849   | 526   | 514   | 580   | 422   | 123   | 136   | 37    | 10    | 22    | 50    | 445   | 500  |
|                                                                                  | PrecMin     | 780    | 124   | 61    | 70    | 7     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 44   |
|                                                                                  | PrecStd     | 310    | 105   | 127   | 142   | 106   | 35    | 28    | 8     | 2     | 5     | 15    | 93    | 118  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 34.6  | 34.3  | 34.2  | 33.3  | 32.8  | 31.6  | 31.2  | 31.6  | 32.7  | 33.8  | 34.5  | 35.5 |
|                                                                                  |             | MCWB   | 27.0  | 26.9  | 26.1  | 25.1  | 24.4  | 23.0  | 21.9  | 20.4  | 21.6  | 23.2  | 25.0  | 26.2 |
|                                                                                  | 2%          | DB     | 33.5  | 33.0  | 33.1  | 32.6  | 31.9  | 30.7  | 30.3  | 30.6  | 31.5  | 32.7  | 33.4  | 34.2 |
|                                                                                  |             | MCWB   | 27.0  | 26.8  | 26.2  | 25.0  | 24.1  | 22.8  | 21.7  | 20.9  | 22.6  | 24.5  | 25.5  | 26.0 |
|                                                                                  | 5%          | DB     | 32.6  | 32.2  | 32.3  | 31.9  | 31.2  | 30.0  | 29.5  | 29.7  | 30.7  | 31.9  | 32.7  | 33.4 |
|                                                                                  |             | MCWB   | 26.8  | 26.6  | 26.2  | 24.8  | 23.9  | 22.6  | 21.5  | 20.9  | 22.9  | 24.7  | 25.5  | 26.0 |
|                                                                                  | 10%         | DB     | 31.8  | 31.4  | 31.5  | 31.3  | 30.5  | 29.2  | 28.6  | 29.0  | 29.9  | 31.2  | 32.2  | 32.7 |
|                                                                                  |             | MCWB   | 26.7  | 26.5  | 26.2  | 24.8  | 23.8  | 22.3  | 21.3  | 21.1  | 22.9  | 24.6  | 25.4  | 26.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 29.2  | 28.8  | 28.8  | 27.7  | 26.6  | 25.7  | 24.6  | 25.2  | 26.1  | 27.4  | 27.9  | 28.3 |
|                                                                                  |             | MCDB   | 32.6  | 31.6  | 31.3  | 29.8  | 29.8  | 28.3  | 28.0  | 27.5  | 28.3  | 30.1  | 31.7  | 31.6 |
|                                                                                  | 2%          | WB     | 28.3  | 28.0  | 28.0  | 26.7  | 25.8  | 24.6  | 23.7  | 24.1  | 25.1  | 26.5  | 27.1  | 27.6 |
|                                                                                  |             | MCDB   | 31.4  | 30.9  | 30.7  | 29.7  | 29.7  | 28.2  | 27.5  | 27.2  | 28.6  | 29.8  | 31.1  | 31.3 |
|                                                                                  | 5%          | WB     | 27.7  | 27.4  | 27.4  | 26.2  | 25.3  | 24.1  | 23.1  | 23.3  | 24.5  | 25.9  | 26.6  | 27.1 |
|                                                                                  |             | MCDB   | 30.8  | 30.5  | 30.4  | 29.7  | 29.2  | 28.2  | 27.3  | 27.1  | 28.4  | 29.8  | 30.7  | 31.1 |
|                                                                                  | 10%         | WB     | 27.1  | 26.9  | 26.8  | 25.7  | 24.8  | 23.5  | 22.5  | 22.6  | 24.1  | 25.3  | 26.1  | 26.7 |
|                                                                                  |             | MCDB   | 30.4  | 30.1  | 29.9  | 29.4  | 28.7  | 27.6  | 26.8  | 26.8  | 28.1  | 29.6  | 30.5  | 30.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.2   | 6.0   | 6.4   | 7.4   | 8.2   | 9.1   | 9.7   | 10.0  | 9.0   | 7.7   | 7.2   | 6.7  |
|                                                                                  |             | MCDBR  | 7.4   | 7.2   | 7.7   | 8.2   | 8.6   | 9.1   | 9.9   | 10.5  | 9.6   | 8.3   | 7.6   | 7.5  |
|                                                                                  |             | MCWBR  | 3.0   | 2.8   | 3.0   | 2.7   | 2.9   | 3.1   | 3.6   | 3.9   | 3.5   | 2.8   | 2.3   | 2.3  |
|                                                                                  | 5% WB       | MCDBR  | 6.8   | 6.6   | 6.8   | 7.1   | 7.3   | 7.9   | 8.6   | 8.5   | 8.1   | 7.3   | 7.2   | 7.0  |
|                                                                                  |             | MCWBR  | 3.8   | 3.4   | 3.7   | 2.9   | 2.9   | 3.2   | 3.7   | 3.9   | 3.4   | 2.8   | 2.7   | 2.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.414  | 0.404 | 0.395 | 0.382 | 0.375 | 0.362 | 0.353 | 0.363 | 0.405 | 0.419 | 0.425 | 0.415 |      |
|                                                                                  | taud        | 2.528  | 2.556 | 2.579 | 2.602 | 2.590 | 2.591 | 2.610 | 2.556 | 2.423 | 2.403 | 2.429 | 2.504 |      |
|                                                                                  | Ebn at Noon | 929    | 932   | 923   | 905   | 880   | 876   | 893   | 910   | 900   | 908   | 913   | 927   |      |
|                                                                                  | Edh at Noon | 112    | 108   | 103   | 96    | 92    | 89    | 89    | 99    | 119   | 125   | 123   | 114   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 5.28   | 5.41  | 5.29  | 5.35  | 5.11  | 4.88  | 5.17  | 5.96  | 6.65  | 6.97  | 6.83  | 5.95  |      |
|                                                                                  | RadStd      | 0.65   | 0.81  | 0.59  | 0.66  | 0.38  | 0.27  | 0.21  | 0.21  | 0.26  | 0.30  | 0.31  | 0.58  |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (4 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page